

**AMENDMENT NO. 1 DECEMBER 2006
TO
IS 11447 : 1985 CODE OF PRACTICE FOR
CONSTRUCTION WITH LARGE PANEL
PREFABRICATES**

(Page 3, clause 0.2) — Insert the following para at the end:

“The various structural schemes are possible using large panel prefabricates. A reference to Part 6 Structural design, Section 7A Prefabricated concrete of SP 7 : 2005 ‘National Building Code of India 2005’ may be made in this regard.”

(Page 3, clause 0.4) — Insert the following clause at the end:

‘0.5 All standards, whether given herein above or cross-referred to in the main text of this standard, are subject to revision. The parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards.’

(Page 7, clause 4.1.2) — Insert the following at the end:

“The structural design of elements shall be required to take care the provisions specified for earthquake resistant design of buildings in IS 1893 (Part 1) : 2002 ‘Criteria for earthquake resistant design of structures: Part 1 General provisions and buildings (*fifth revision*)’.”

(Page 9 clause 4.1.7) — Insert the following at the end:

‘4.1.8 The panels shall be designed and fabricated considering the design and laying requirements of various services, such as electrical, public health, mechanical and HVAC for the building.’

(Page 13, clause 4.5.1) — Substitute the following for the existing clause:

‘4.5.1 The walls may be designed as unreinforced or reinforced cement concrete. It is recommended to design walls as plain concrete with secondary reinforcement for production and erection purpose only which is not taken into account while checking the safety of wall. For walls, concrete of grade not less than M 20 shall be used.’

4.5.1.1 *External walls*

The external walls, their horizontal and vertical joints/junctions and junctions at doors/windows shall satisfy the following requirements:

- a) They should provide resistance to loads either vertical or horizontal, if required; this depends on the layout of the building;
- b) The fire resistance, thermal comfort and natural lighting shall be in accordance with the requirements of relevant Indian Standards; and
- c) They should provide adequate protection against weather to the building and its occupants.

External walls may be classified as homogenous walls and non-homogenous walls. They may be load bearing as well as non-load bearing walls. The thickness of the walls shall depend on the minimum structural needs as well as the thickness needed at joints to ensure water tightness. Non-homogenous walls may be sandwich panel type, composed of two leaves separated by a layer of insulation, or they may be framed infill type. In sandwich panels, the external and internal walls are both of concrete, appropriately interconnected through the central layer of insulation. All walls shall be designed and detailed to adequately resist the effects of temperature and shrinkage.

In infill walls, where there is a perimeter frame enclosing masonry infills, the frame shall be structurally adequate to carry all imposed loads. Intermediate ribs may be introduced to improve the structural action

For sandwich panels, the minimum thickness of external concrete layer shall be 40 mm when its movement is constrained. When it has free movement, its minimum thickness shall be 60 mm and it should be connected with internal layer by means of suitable clips allowing free movement.

Different shapes of panels are possible for external walls. Windows may be cast into the panel itself, or they may be fixed at site.

4.5.1.2 *Internal walls*

Internal structural walls may be homogenous walls of cement concrete (reinforced or unreinforced) or light weight concrete. In case steel is required across panel joints, the joint should be designed to provide continuity in bars.

(Page 13, clause 4.5.2) — Insert the following at the end:

‘However the thickness of walls shall meet the requirements for earthquake resistance in accordance with IS 1893 (Part 1) : 2002.’

(Page 24, clause 4.9.5) — Insert the following at the end:

‘4.9.6 While using large panel prefabricates, the structural integrity of the whole structure has to be ensured. For various applications, for ensuring diaphragm action, the provisions specified in IS 1893 (Part 1) : 2002 and IS 4326 : 1993 ‘Code of practice for earthquake resistant design and construction of buildings (second revision)’ shall be followed.’

(CED 51)

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